Estimation of Linear Regression

M514: Econometrics II

Sangmyung Ha

Indiana University

Spring 2024

Estimation of linear projection coefficient

We estimate β as the minimizer of the sample analog of the mean squared error $\mathbb{E}\left[(Y - X'\beta)^2\right]$. Notice that

$$\begin{aligned}
\hat{\mathbb{E}}\left[(Y - X'\beta)^2\right] &= \frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i'\beta)^2 \\
&= \frac{1}{n}(\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta)
\end{aligned}$$

where

$$\mathbf{X} = \begin{pmatrix} X_1' \\ \vdots \\ X_n' \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}$$

Estimation of linear projection coefficient

We want to find $\beta \in \mathbb{R}^k$ which solves

$$\max_{\beta \in \mathbb{R}^k} \frac{1}{n} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) \tag{1}$$

The FOC is given by

$$\begin{aligned}
0 &= \frac{\partial}{\partial \beta} \left\{ \frac{1}{n} (\mathbf{Y} - \mathbf{X}\beta)'(\mathbf{Y} - \mathbf{X}\beta) \right\} \\
&= \frac{1}{n} \frac{\partial}{\partial \beta} \left\{ \mathbf{Y}'\mathbf{Y} - 2\beta'\mathbf{X}'\mathbf{Y} + \beta'\mathbf{X}'\mathbf{X}\beta \right\} \\
&= \frac{1}{n} \left(-2\mathbf{X}'\mathbf{Y} + 2\mathbf{X}'\mathbf{X}\beta \right) \\
\implies \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}
\end{aligned}$$

Fitted values

The *fitted values* $\hat{\mathbf{Y}} \in \mathbb{R}^n$ are given by

$$\begin{aligned}\hat{\mathbf{Y}} &= \mathbf{X}\hat{\beta} \\ &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}\end{aligned}$$

Notice that $\hat{\mathbf{Y}}$ is given by projecting $\mathbf{Y}$ onto column space of $\mathbf{X}$. Projecting $\mathbf{Y} \in \mathbb{R}^n$ onto column space of $\mathbf{X}$ means finding an element in column space of $\mathbf{X}$ such that distance between $\mathbf{Y}$ and that element is minimized. This is equivalent to finding a linear combination of columns of $\mathbf{X}$ such that the distance between the linear combination and $\mathbf{Y}$ is minimized. This is exactly what (1) tries to solve.

Projection residuals

The *projection residuals* $\hat{\mathbf{e}} \in \mathbb{R}^n$ are given by

$$\begin{aligned}\hat{\mathbf{e}} &= \mathbf{Y} - \hat{\mathbf{Y}} \\ &= \mathbf{Y} - \mathbf{X}\hat{\beta} \\ &= \mathbf{Y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}\end{aligned}$$

Since $\hat{\mathbf{e}}$ is the residual from orthogonal projection onto columns space of $\mathbf{X}$, $\hat{\mathbf{e}}$ is orthogonal to each columns of $\mathbf{X}$, i.e.,

$$\begin{aligned}\mathbf{X}'\hat{\mathbf{e}} &= \mathbf{X}'(\mathbf{Y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}) \\ &= \mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\ &= \mathbf{X}'\mathbf{Y} - \mathbf{X}'\mathbf{Y} \\ &= 0\end{aligned}$$

R^2

The *r-squared* measures how well $\mathbf{X}$ can explain variations of $\mathbf{Y}$. The r-squared is defined by

$$R^2 = 1 - \frac{(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta})}{(\mathbf{Y} - \bar{\mathbf{Y}})'(\mathbf{Y} - \bar{\mathbf{Y}})}$$

where $\hat{\beta}$ minimizes $(\mathbf{Y} - \mathbf{X}\hat{\beta})'(\mathbf{Y} - \mathbf{X}\hat{\beta})$. Including additional variables in X will always increase the R^2, unless the column vector with sample values of the additional variable is in the column space of $\mathbf{X}$, in which case R^2 will not change. If we include enough variables so that $k \geq n$, $\mathbf{Y}$ will in general be in the column space of $\mathbf{X}$ and R^2 will be one.

Linear Regression Model

1. **Assumption 1**
   $(X_i, Y_i), i = 1, \ldots, n$ are iid.
2. **Assumption 2**
   The variables (X, Y) satisfy the linear regression equation
   $$Y = X'\beta + e$$
   $$\mathbb{E}[e|X] = 0$$
3. **Assumption 3**
   The error terms are conditionally homoskedastic, i.e.,
   $$\mathbb{E}[e^2|X] = \sigma^2(X) = \sigma^2$$
   is independent of X

Bias of $\hat{\beta}$

The matrix notation for the linear regression model given in Assumption 2 is

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{e}$$

Under Assumption 1 and 2, $\hat{\beta}$ is *unbiased*, i.e., we have

$$\begin{aligned}
\mathbb{E}[\hat{\beta}|\mathbf{X}] &= \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}|\mathbf{X}] \\
&= \mathbb{E}[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e})|\mathbf{X}] \\
&= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}[\mathbf{e}|\mathbf{X}] \\
&= \beta
\end{aligned}$$

due to the *conditional mean zero assumption*, i.e., $\mathbb{E}[e|\mathbf{X}] = 0$. Here, we assume that the inverse of $\mathbf{X}'\mathbf{X}$ exists, which is called the assumption of *no perfect multicollinearity*.

Conditional variance of $\hat{\beta}$

The conditional variance of $\hat{\beta}$ is given by

$$\mathsf{Var}(\hat{\beta}|\mathbf{X}) = \mathbb{E}\left[\left(\hat{\beta} - \mathbb{E}[\hat{\beta}|\mathbf{X}]\right)\left(\hat{\beta} - \mathbb{E}[\hat{\beta}|\mathbf{X}]\right)' |\mathbf{X}\right]$$

From the previous subsection, we have $\mathbb{E}[\hat{\beta}|\mathbf{X}] = \beta$. From

$$\begin{aligned}
\hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e}) \\
&= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}
\end{aligned}$$

we have

$$\begin{aligned}
\mathsf{Var}(\hat{\beta}|\mathbf{X}) &= \mathbb{E}\left[\left(\hat{\beta} - \beta\right)\left(\hat{\beta} - \beta\right)' |\mathbf{X}\right] \\
&= \mathbb{E}\left[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}\mathbf{e}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}|\mathbf{X}\right] \\
&= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}\left[\mathbf{e}\mathbf{e}'|\mathbf{X}\right]\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}
\end{aligned}$$

Conditional variance of $\hat{\beta}$

Due to Assumption 3, we have $\mathbb{E}\left[\mathbf{e}\mathbf{e}'|\mathbf{X}\right] = \sigma^2 I_n$, which gives

$$\begin{aligned}\text{Var}(\hat{\beta}|\mathbf{X}) &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbb{E}\left[\mathbf{e}\mathbf{e}'|\mathbf{X}\right]\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}\end{aligned}$$

Thus, the variance of $\hat{\beta}_j$ increases as σ^2 increases and j-th diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$ increases.

Conditional variance of $\hat{\beta}$

Let the spectral representation of $\mathbf{X}'\mathbf{X}$ be given by

$$\mathbf{X}'\mathbf{X} = UDU'$$

Then, we have

$$(\mathbf{X}'\mathbf{X})^{-1} = UD^{-1}U'$$

In general, diagonal elements of $(\mathbf{X}'\mathbf{X})^{-1}$ tends to increase as the eigenvalues of $\mathbf{X}'\mathbf{X}$ decreases. If some of eigenvalues of $\mathbf{X}'\mathbf{X}$ are very close to zero, i.e., $\mathbf{X}'\mathbf{X}$ is near singular, elements of $(\mathbf{X}'\mathbf{X})^{-1}$ will tend to explode. This problem is called *multicollinearity*. This can happen if some of the columns of $\mathbf{X}$ can be very closely approximated by some linear combination of all other columns.
Under Assumptions 1, 2, and 3, the OLS estimator $\hat{\beta}$ has the lowest variance among all linear unbiased estimators. This result is called the *Gauss-Markov Theorem*.

Weighted MSE

Gauss-Markov theorem compares the OLS estimator only to the class of unbiased estimators. Although OLS is best among all linear unbiased estimators, we might be able to find a better estimator if we consider biased estimators. When comparing two estimators that may have different biases, it's important to consider a criterion beyond variance alone. The conditional *weighted MSE* given $\mathbf{X}$ for an estimator $\tilde{\beta}$ is defined as

$$\text{wMSE}(\tilde{\beta}|\mathbf{X}) = \mathbb{E}[(\tilde{\beta} - \beta)' W((\tilde{\beta} - \beta))|\mathbf{X}]$$

where W is $k \times k$ dimensional weighting matrix.

Weighted MSE

Usually we consider $W = V^{-1}$ where V is the variance of the OLS estimator $\hat{\beta}$. The weighted MSE of a generic $\tilde{\beta}$ is given by

$$
\begin{aligned}
\text{wMSE}(\tilde{\beta}|\mathbf{X}) &= \mathbb{E}[(\tilde{\beta} - \beta)' W (\tilde{\beta} - \beta)|\mathbf{X}] \\
&= \mathbb{E}\left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] + \mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}] + \mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right) |\mathbf{X}\right] \\
&= \mathbb{E}\left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' W \left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right) |\mathbf{X}\right] + \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right) \\
&\quad + 2\mathbb{E}\left[\left(\tilde{\beta} - \mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}] - \beta\right) |\mathbf{X}\right]
\end{aligned}
$$

Weighted MSE

Here, we have

$$
\begin{aligned}
\mathbb{E}\left[\left(\tilde{\beta}-\mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}]-\beta\right)|\mathbf{X}\right] &= \mathbb{E}\left[\left(\tilde{\beta}-\mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)'|\mathbf{X}\right] W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}]-\beta\right) \\
&= \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}]-\mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}]-\beta\right) \\
&= 0
\end{aligned}
$$

Thus, the weighted MSE decomposes as

$$
\text{wMSE}(\tilde{\beta}|\mathbf{X}) = \mathbb{E}\left[\left(\tilde{\beta}-\mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)' W \left(\tilde{\beta}-\mathbb{E}[\tilde{\beta}|\mathbf{X}]\right)|\mathbf{X}\right] + \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}]-\beta\right)' W \left(\mathbb{E}[\tilde{\beta}|\mathbf{X}]-\beta\right)
$$

Weighted MSE

For the OLS estimator $\hat{\beta}$, we have

$$\begin{aligned}
\text{wMSE}(\hat{\beta}|\mathbf{X}) &= \mathbb{E}\left[\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)' W \left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)\Big|\mathbf{X}\right] \\
&= \text{tr}\left(\mathbb{E}\left[\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)' W \left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)\Big|\mathbf{X}\right]\right) \\
&= \mathbb{E}\left[\text{tr}\left(\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)' W \left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)\right)\Big|\mathbf{X}\right] \\
&= \mathbb{E}\left[\text{tr}\left(W\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)'\right)\Big|\mathbf{X}\right] \\
&= \text{tr}\left(W\mathbb{E}\left[\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)\left(\hat{\beta}-\mathbb{E}[\hat{\beta}|\mathbf{X}]\right)'\Big|\mathbf{X}\right]\right) \\
&= \text{tr}\left(WV\right) \\
&= k
\end{aligned}$$

assuming that $W = V^{-1}$ had been used.